Event-Based Menu Recommendation Using Sentiment Analysis of Food Preferences

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Introduction

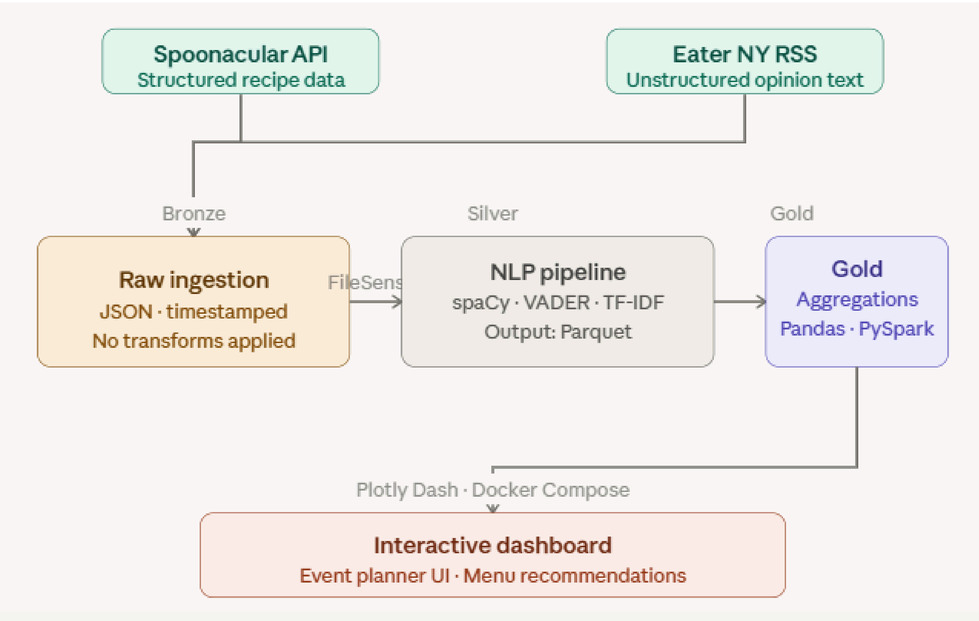
Event-based menu planning in highly competitive environments like **New York City** is a complex task that requires balancing **diverse guest preferences** and **current food trends**. However, most decisions are still based on **intuition** rather than **data-driven insights**. Previous work in sentiment analysis and recommender systems has shown that combining **structured data** with **textual opinions** can improve recommendation quality. The main challenge lies in integrating heterogeneous data sources and extracting meaningful insights from unstructured text.

Goal

The goal of this project is to design a system that answers: How can **public sentiment** and **structured culinary data** be combined to improve event-based menu recommendations?
The expected outcome is an end-to-end data pipeline that generates data-driven dish recommendations tailored to **specific event contexts**.

Proposed Solution

We propose a data pipeline based on the Medallion Architecture that integrates structured data from the Spoonacular API and unstructured text from the Eater NY RSS feed. Data is ingested into a Bronze layer, processed with NLP techniques (sentiment analysis using VADER, entity recognition with spaCy, and keyword extraction) in the Silver layer, and aggregated into insights in the Gold layer. The system is orchestrated using Apache Airflow and visualized through an interactive dashboard built with Plotly Dash.



Experiments

We conducted initial experiments to validate the data ingestion and integration process. A sample of 20 records per source was stored in the Bronze layer to evaluate consistency and usability.
Several data quality issues were identified, especially in the unstructured source, requiring preprocessing before applying NLP techniques.

Source	Issue Type	Description
Eater NY	HTML Noise	Requires cleaning (HTML entities, text noise)
Eater NY	Boilerplate Text	Non-relevant text must be removed
Spoonacular	Nested JSON	Requires flattening
Spoonacular	Complex Strings	Ingredient parsing needed

Results

The results validate the **data ingestion process**, with 20 records per source successfully collected and stored, confirming their **usability.**

- Both sources provided **valid data.**
- Combines **structured and unstructured data.**
- Supports **scalable integration.**
- Limited by **small dataset** and no **NLP processing yet.**

Conclusions

This work demonstrates the feasibility of a **data-driven** approach to **event-based menu recommendation** by integrating structured and unstructured data. The system successfully validates **data ingestion** and establishes a foundation for **sentiment analysis** and **trend detection**, supporting more informed menu decisions.

Bibliography

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